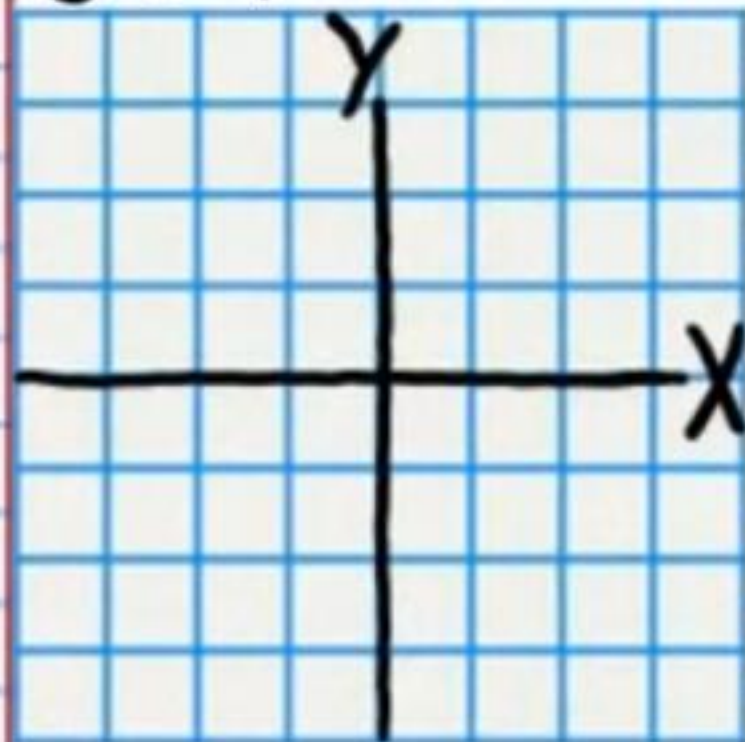


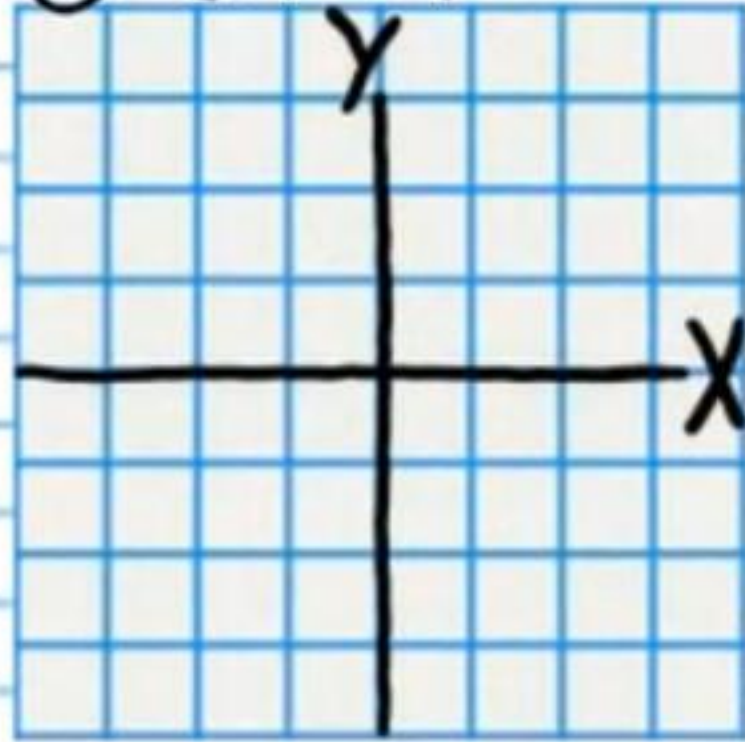
The Complex Plane and Graphing Polar Equations Using Radian Units (Homework)

Draw the points described by the polar coordinates below. Then convert the coordinates so that each angle measure is written with radian units rather than degree units.

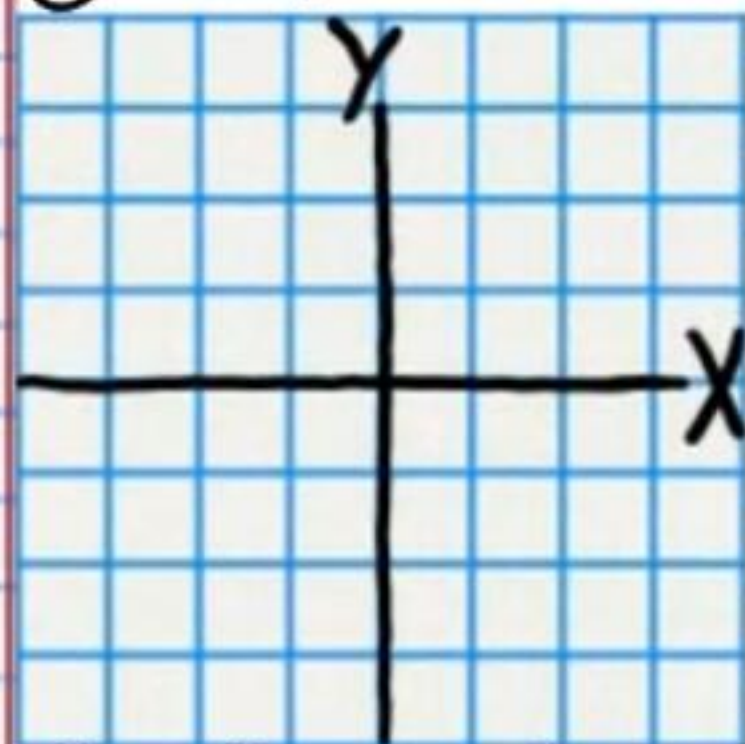
① $(2, 225^\circ)$



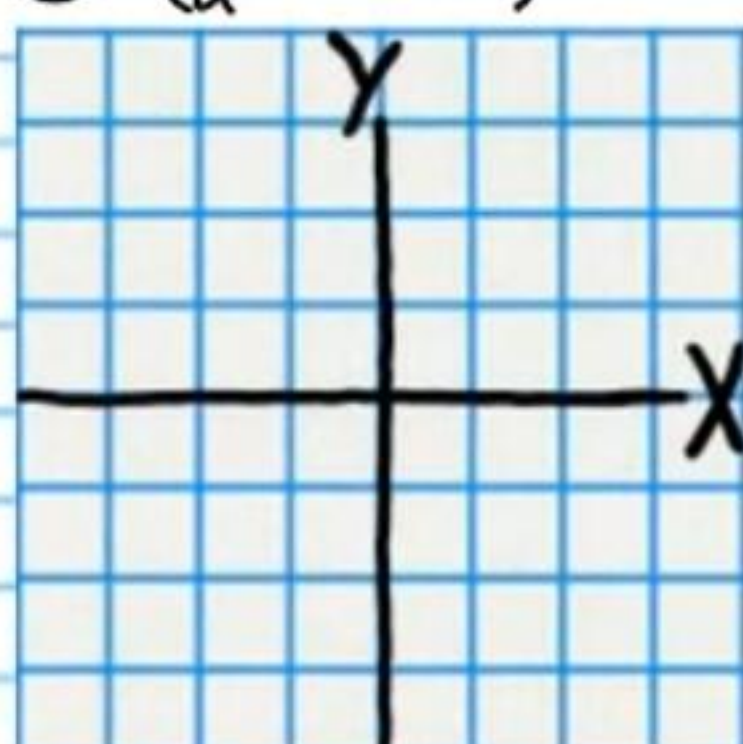
② $(3, 90^\circ)$



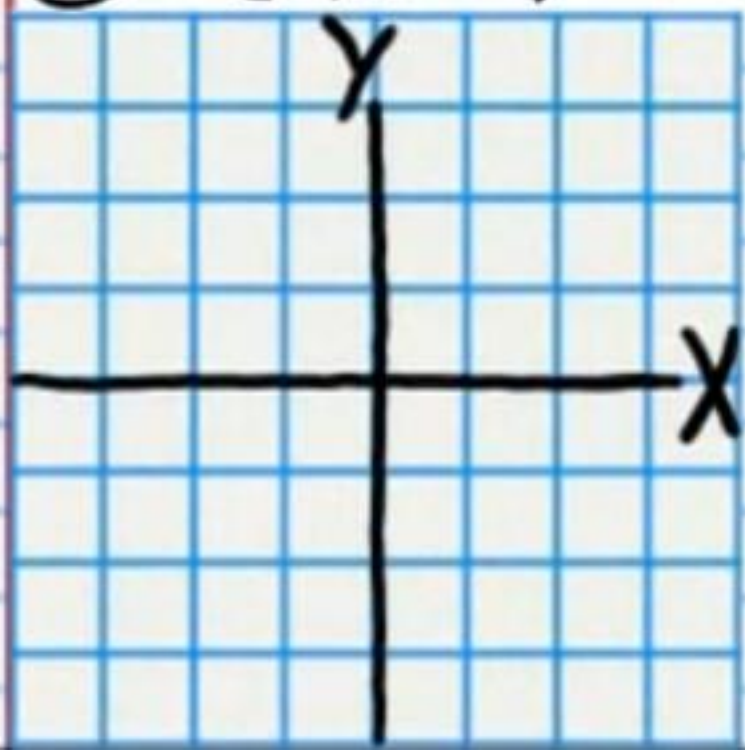
③ $(-1, -150^\circ)$



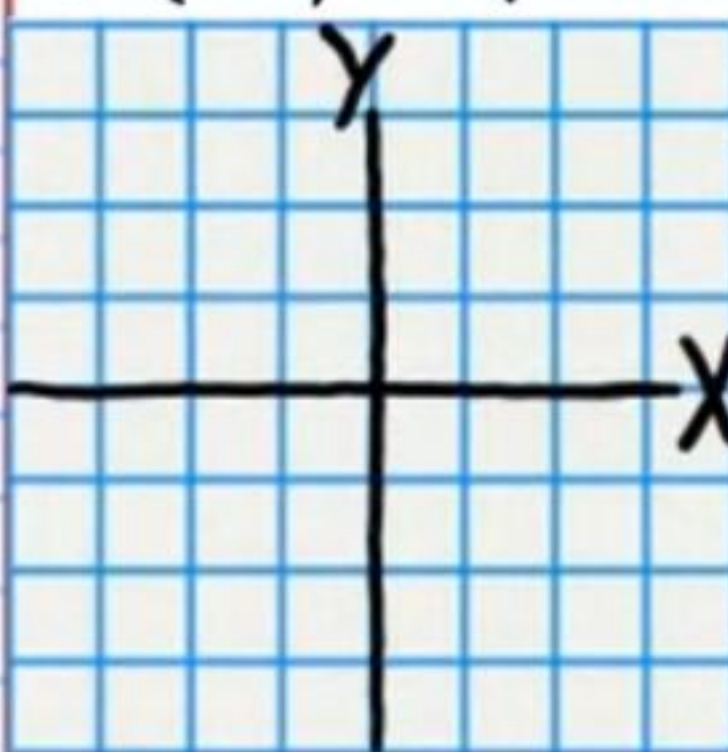
④ $(\frac{1}{2}, -270^\circ)$



⑤ $(3, 248^\circ)$



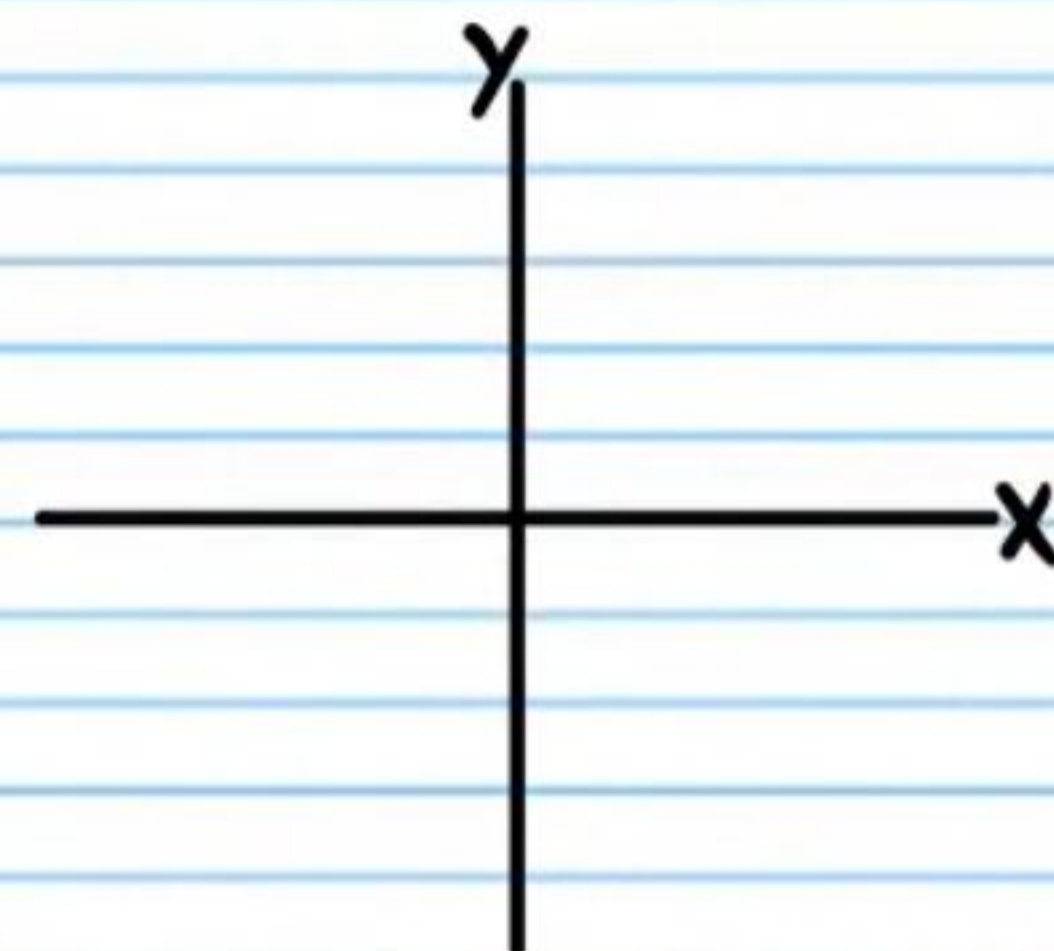
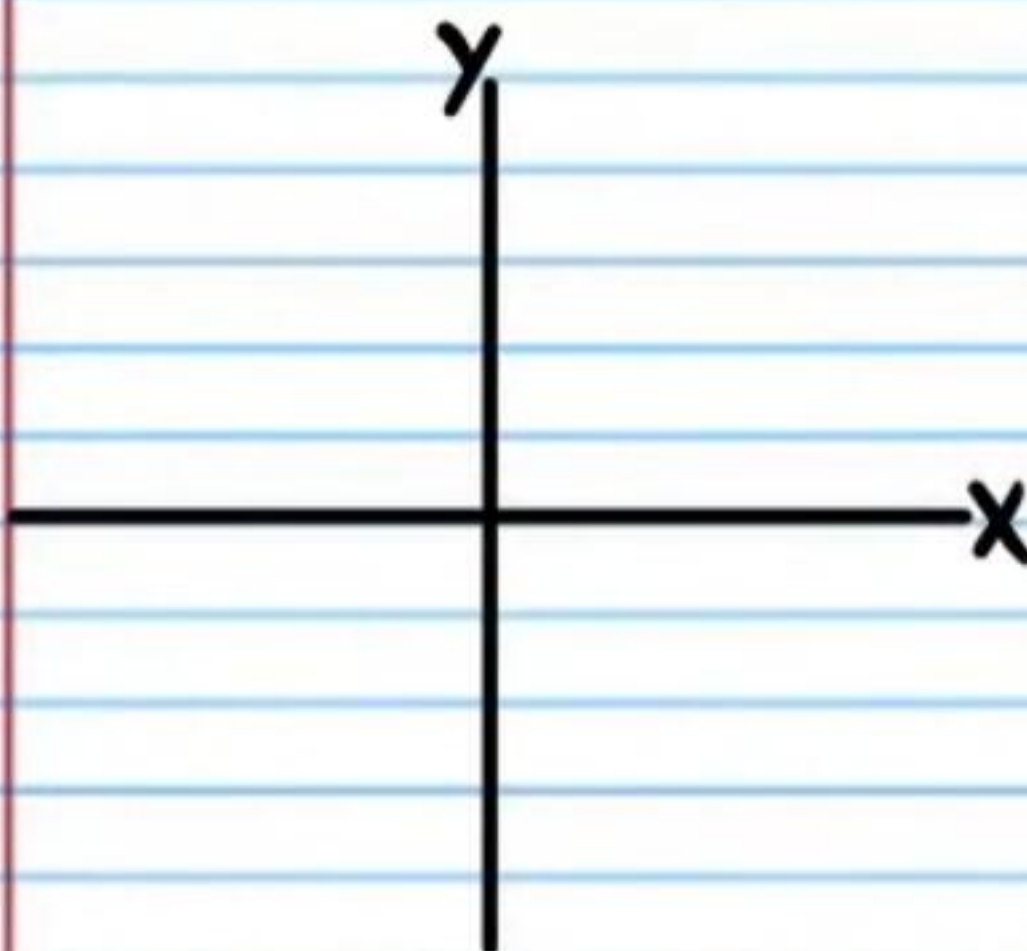
⑥ $(-2, 140^\circ)$



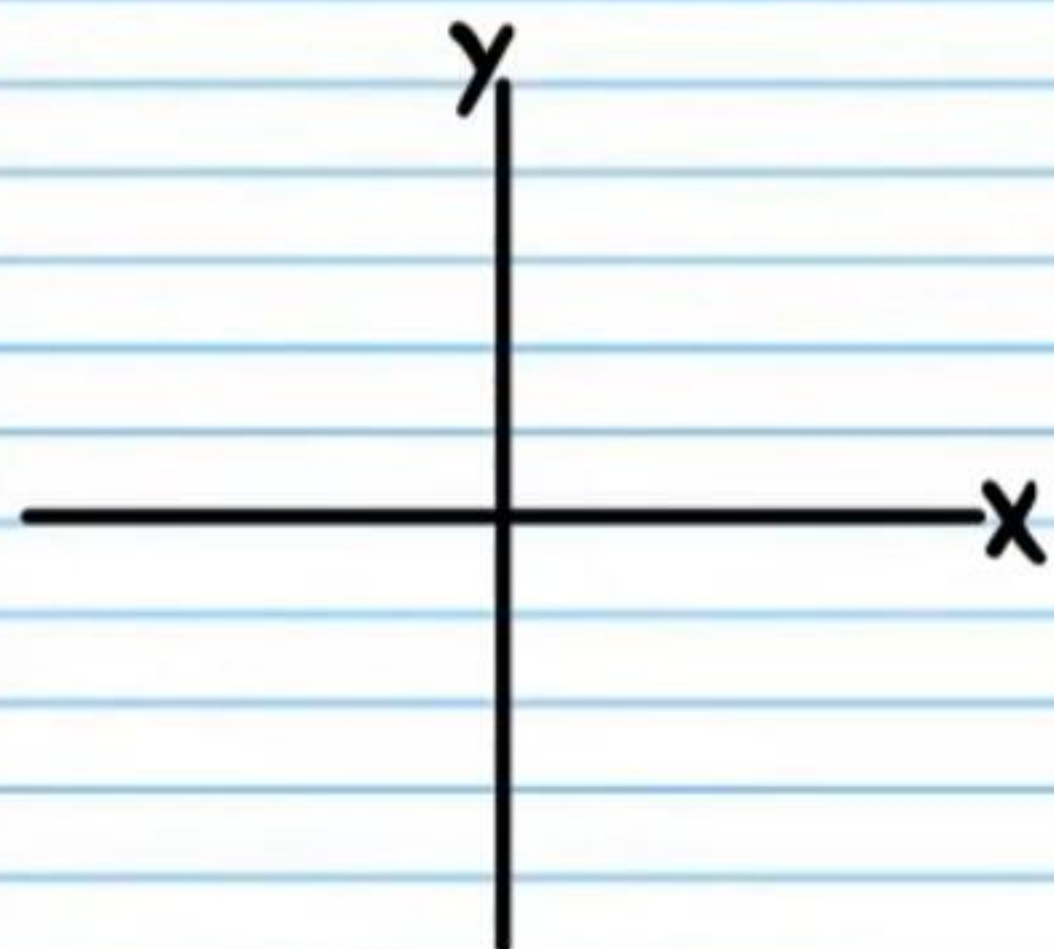
Find the values of the expressions below *without a calculator*.

⑦ $\sin\left(\frac{\pi}{6}\right)$

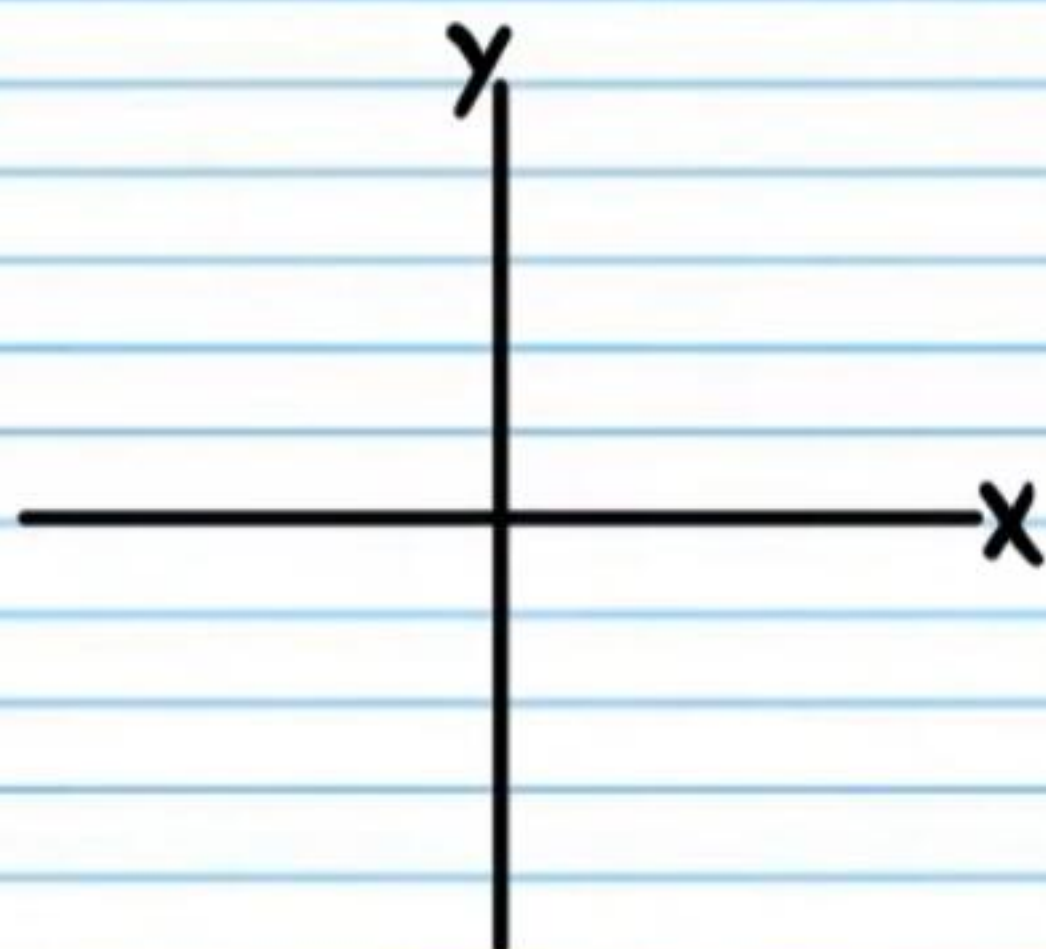
⑧ $\cos\left(\frac{5\pi}{4}\right)$



⑨ $\sec\left(\frac{2\pi}{3}\right)$



⑩ $\cot\left(\frac{7\pi}{4}\right)$



⑪ $\sin\left(\frac{\pi}{2}\right)$

⑫ $\cos\left(\frac{3\pi}{2}\right)$

⑬ $\sin(0)$

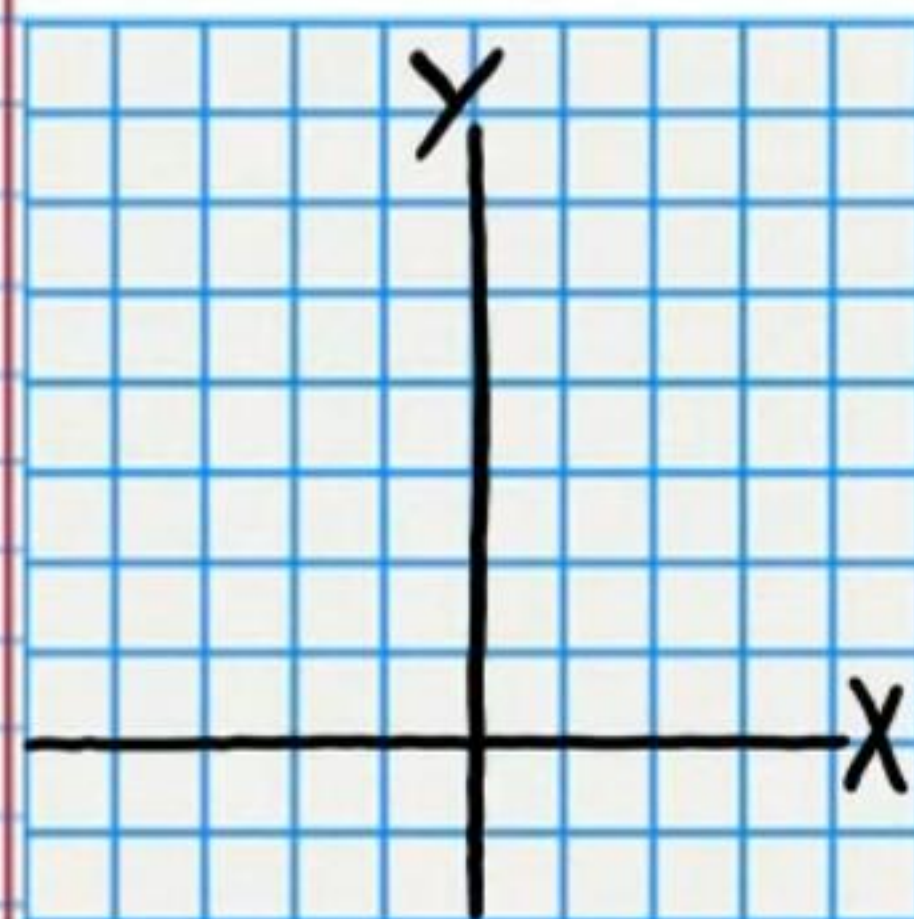
⑭ $\cos\left(-\frac{\pi}{2}\right)$

Graph the following equations using radian units. Write the coordinates of at least 4 points. Use a graphing calculator to draw the other parts of the plot.

⑮ $r = 3 + 2\sin\theta$

1st Point

2nd Point



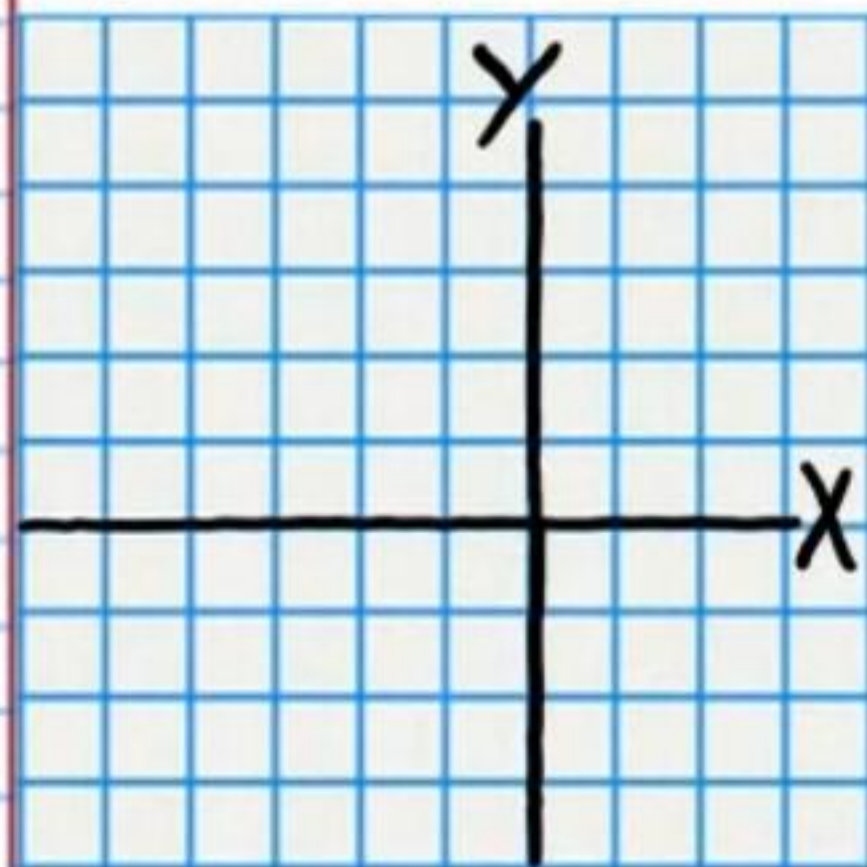
3rd Point

4th Point

⑩ $r = \frac{4}{2-3\cos\theta}$

1st Point

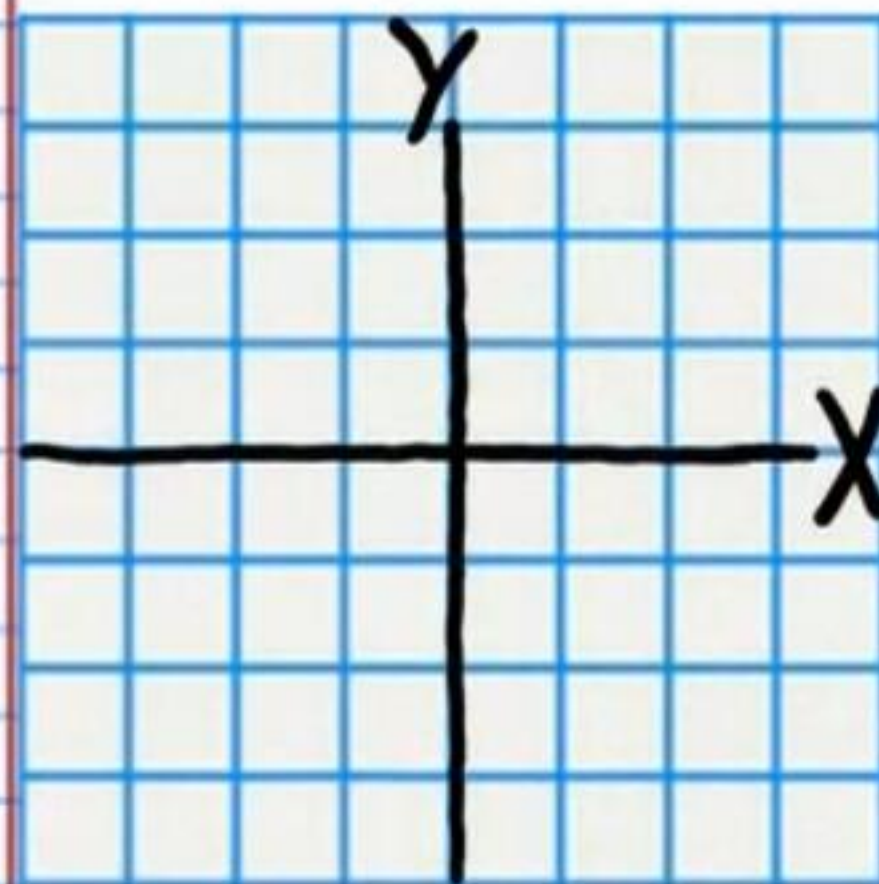
2nd Point



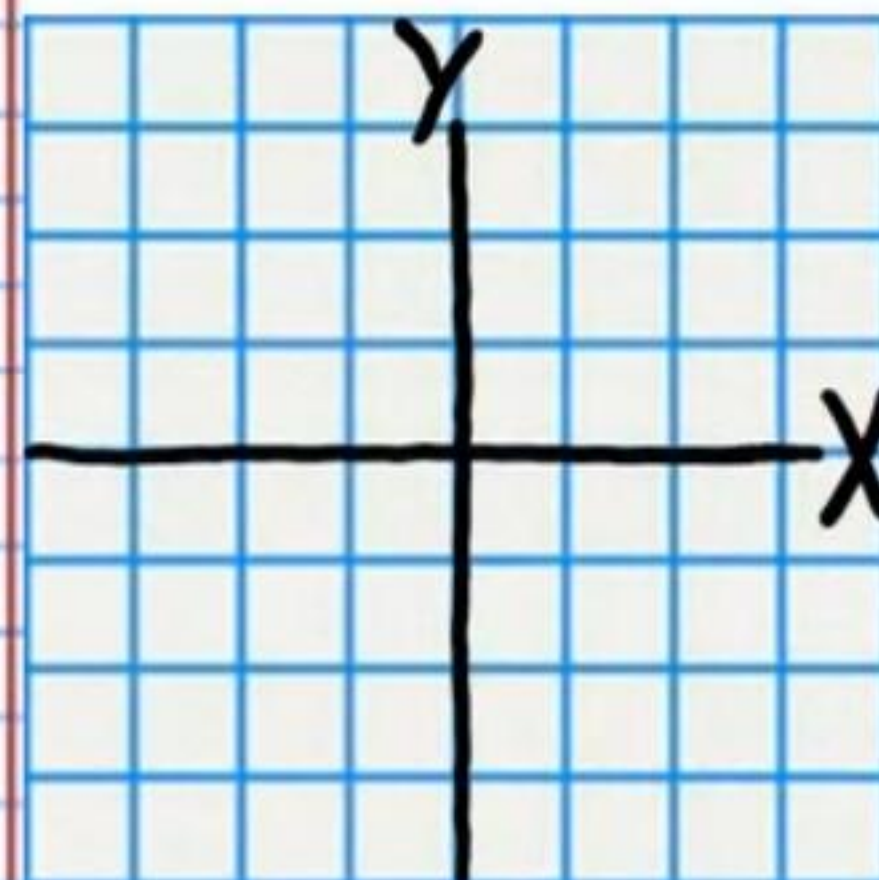
3rd Point

4th Point

⑪ $\theta = \frac{\pi}{6}$



⑫ $\theta = -\frac{3\pi}{8}$



Simplify the expressions below.

• (19) $(8-3i)+(-4-9i)$

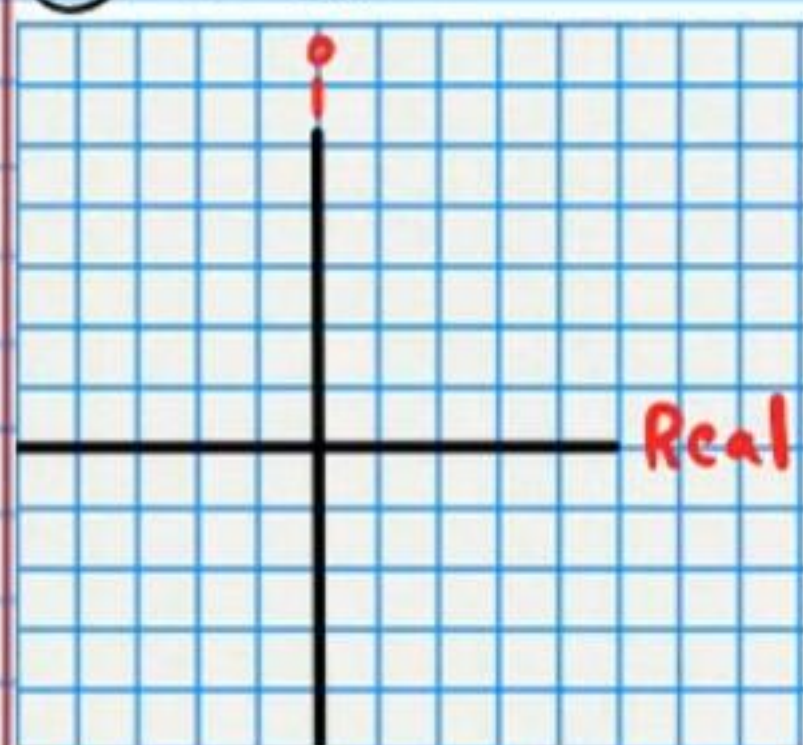
(20) $(10+3i)-(-8+2i)$

(21) $i(11-4i)$

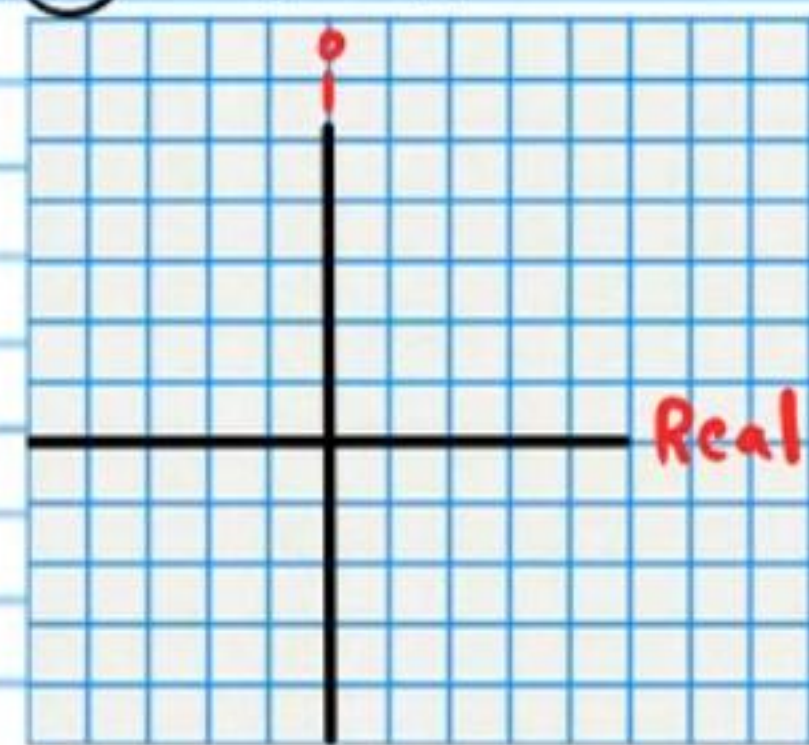
(22) $(7-3i)(4-i)$

Graph the following complex numbers.

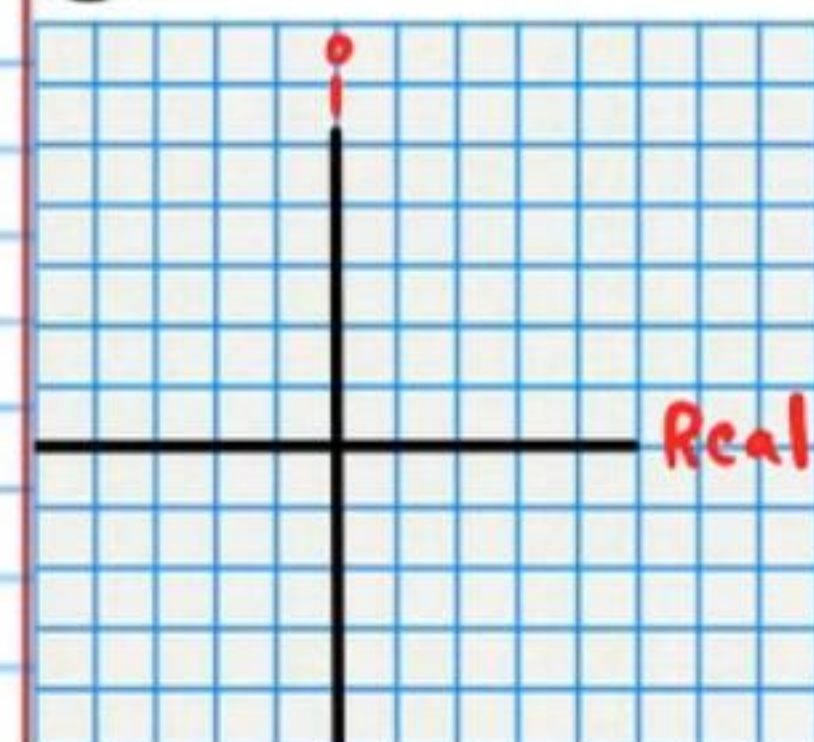
• (41) $1-2i$



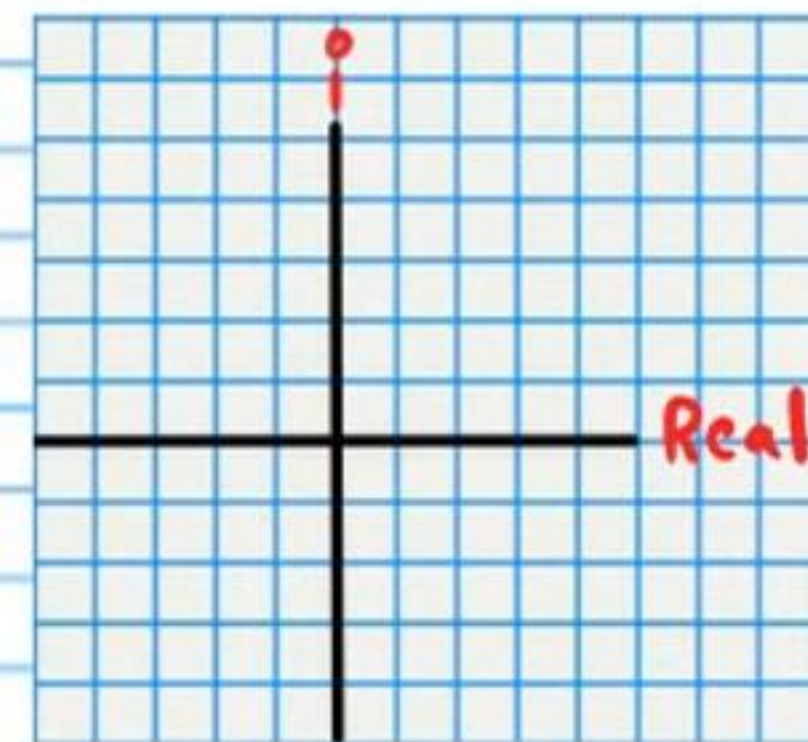
(42) $-4+4i$



(43) $-3-2i$

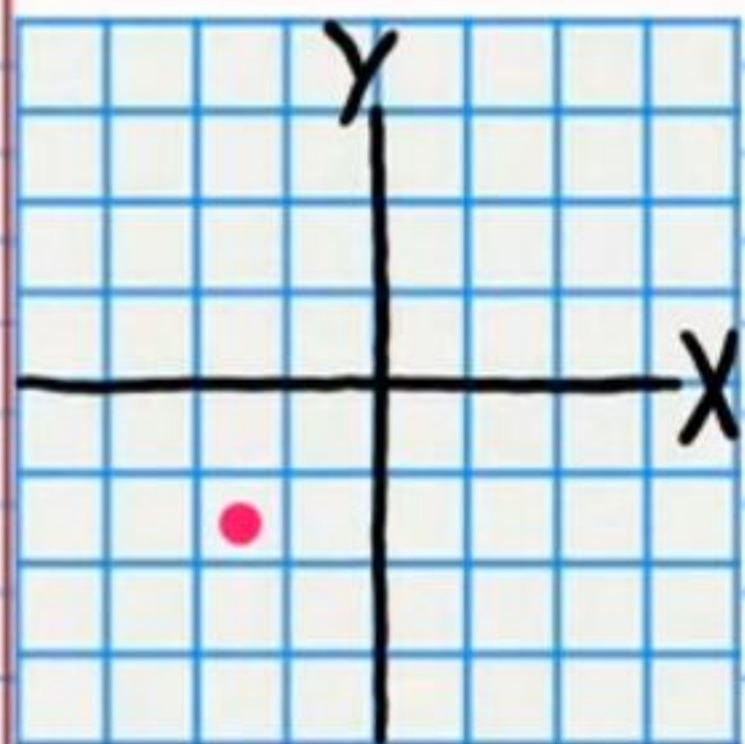


(44) $1+i$

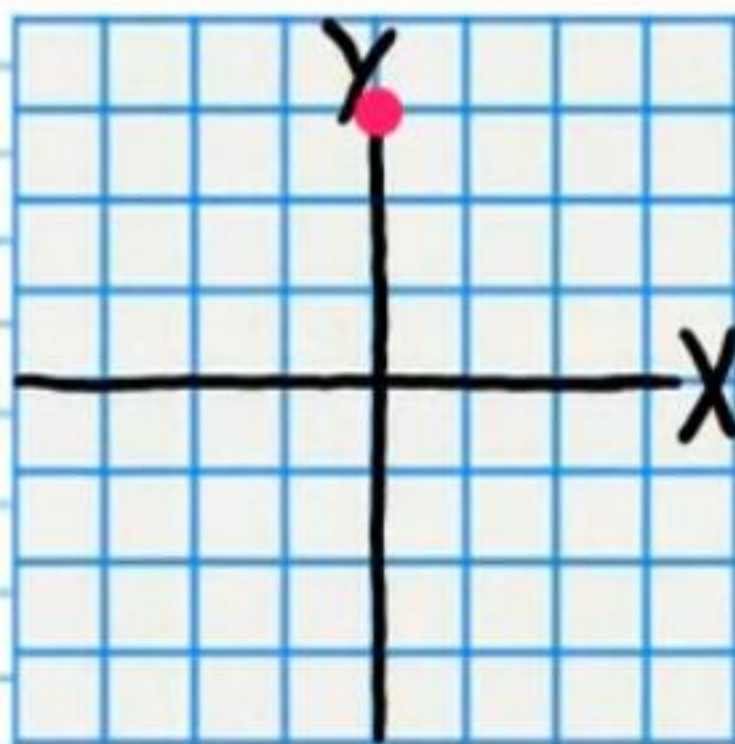


Homework Answers

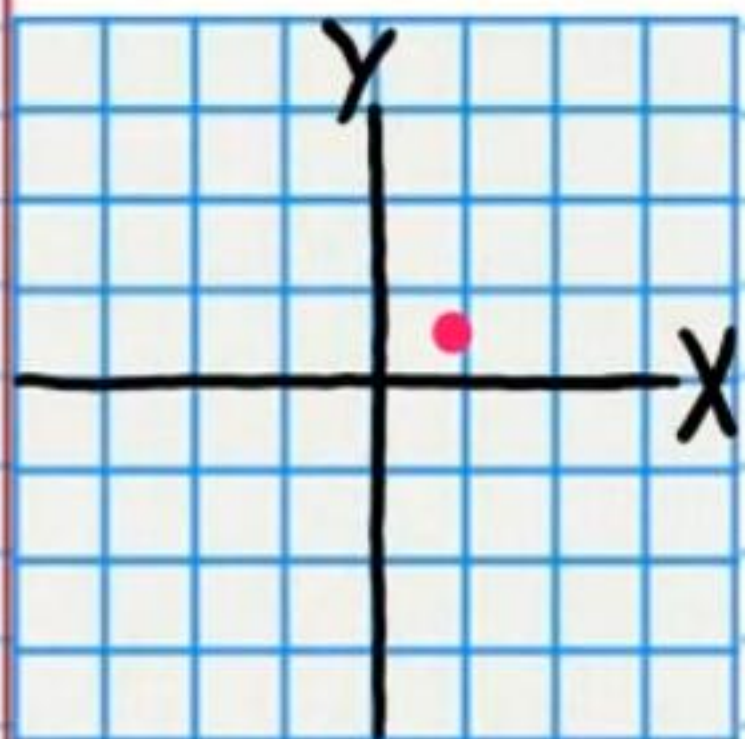
① $(2, \frac{5\pi}{4})$



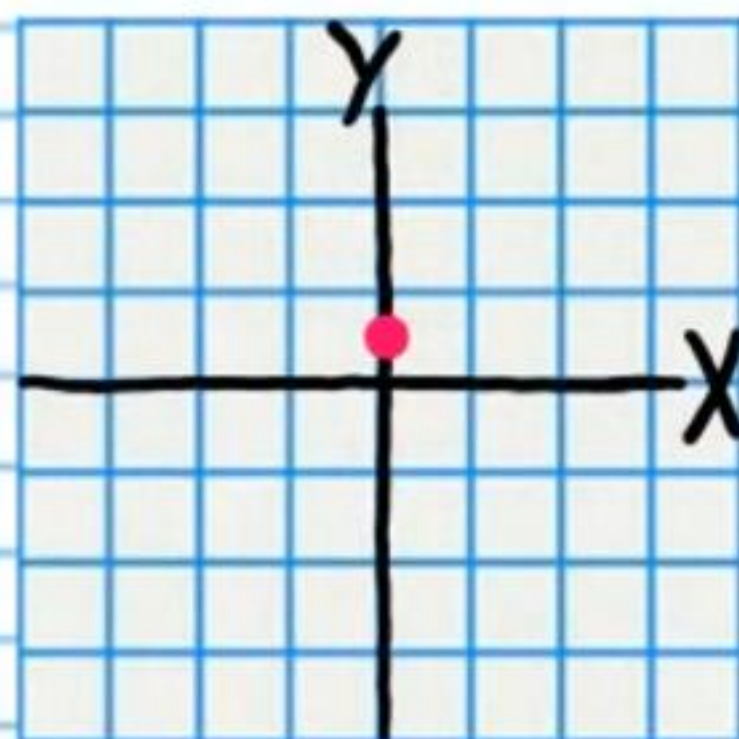
② $(3, \frac{\pi}{2})$



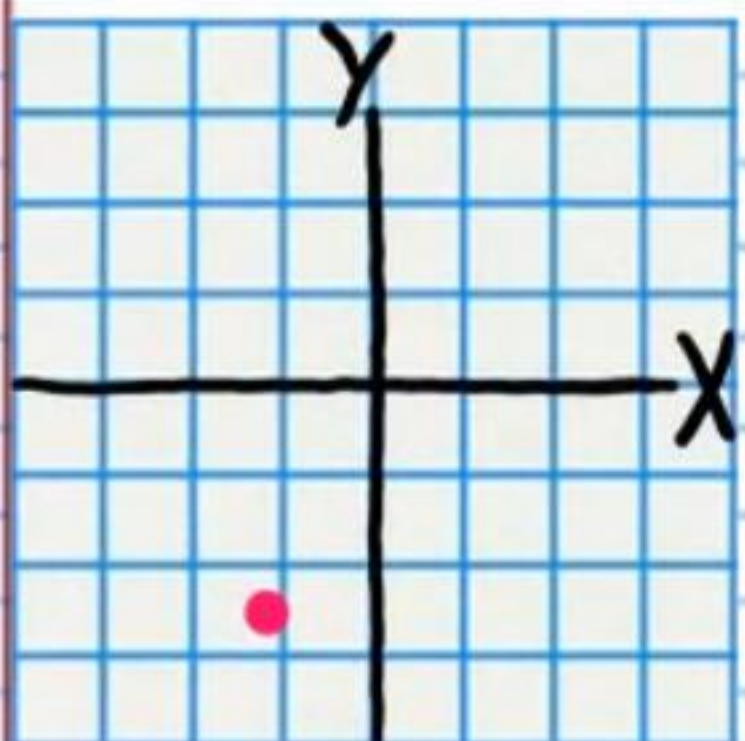
③ $(-1, -\frac{5\pi}{6})$



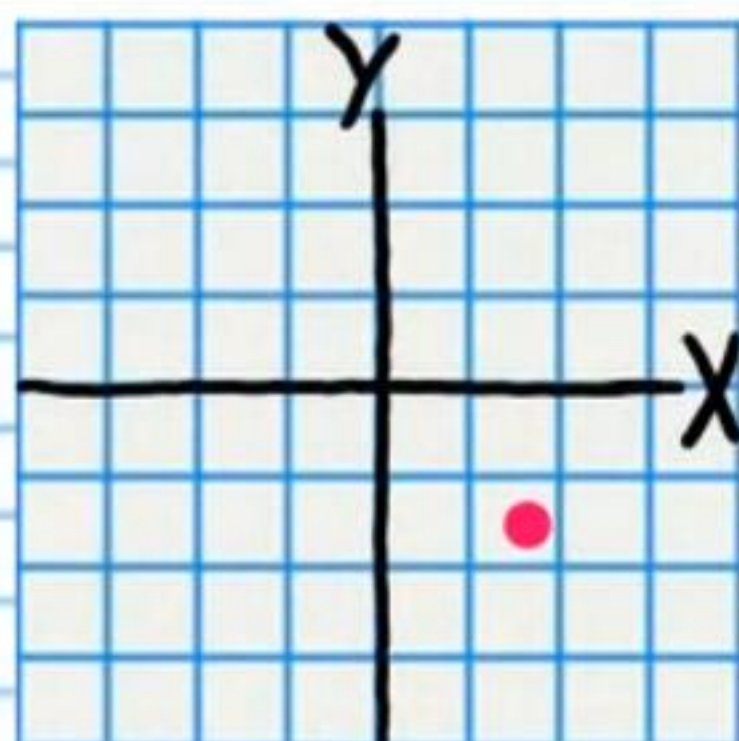
④ $(\frac{1}{2}, -\frac{3\pi}{2})$



⑤ $(3, \frac{62\pi}{45})$



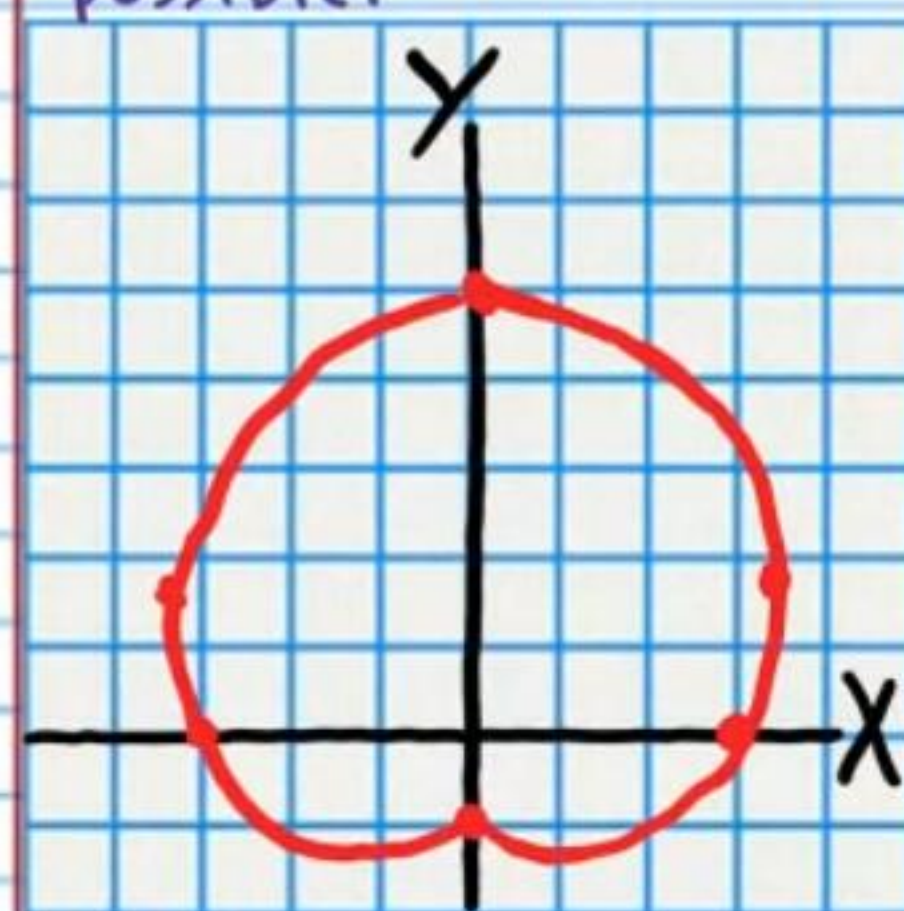
⑥ $(-2, \frac{7\pi}{9})$



⑦ $\frac{1}{2}$ ⑧ $-\frac{\sqrt{2}}{2}$ ⑨ -2 ⑩ -1 ⑪ 1 ⑫ 0

⑬ 0 ⑭ 0

⑮ * Other points are possible.



1st Point

$(3, 0)$

2nd Point

$(5, \frac{\pi}{2})$

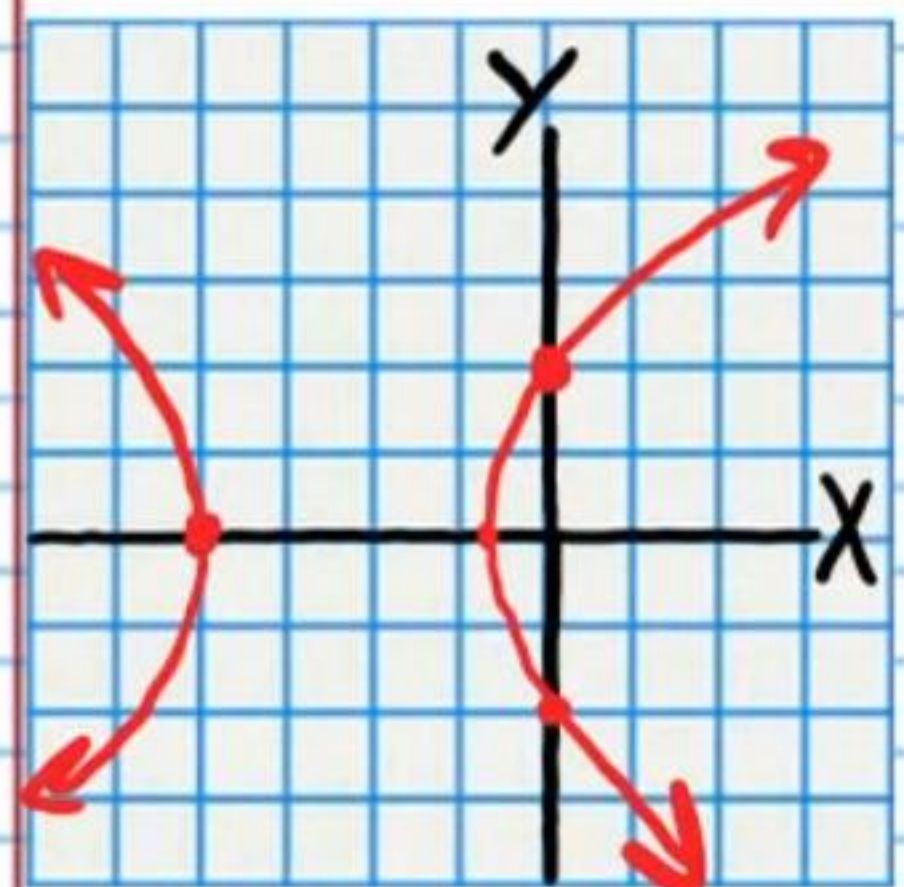
3rd Point

$(1, -\frac{\pi}{2})$

4th Point

$(4, \frac{\pi}{6})$

⑯ * Other points are possible.



1st Point

$(-4, 0)$

2nd Point

$(2, \frac{\pi}{2})$

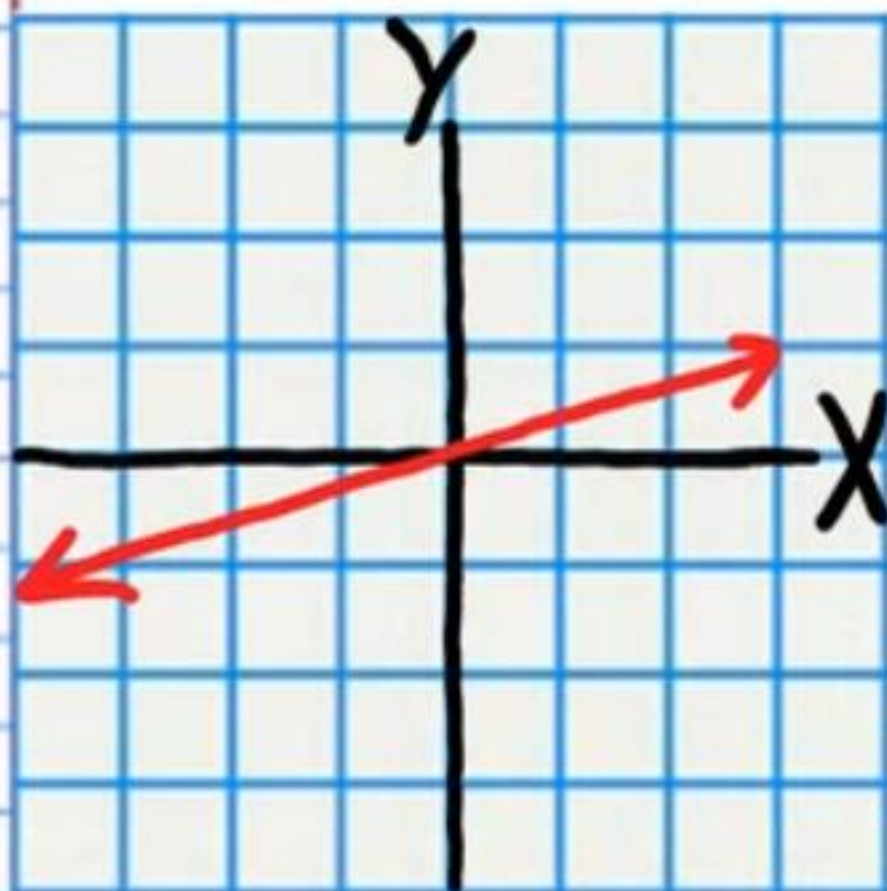
3rd Point

$(2, -\frac{\pi}{2})$

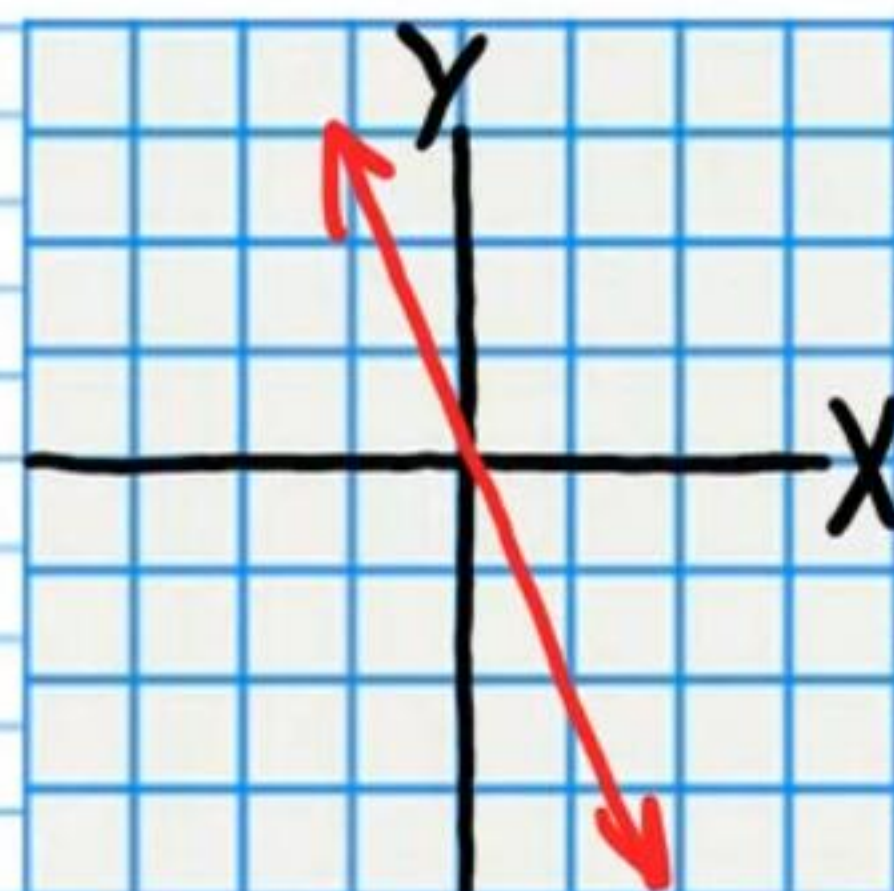
4th Point

$(\frac{4}{5}, \pi)$

⑪ $\theta = \frac{\pi}{6}$



⑫ $\theta = -\frac{3\pi}{8}$



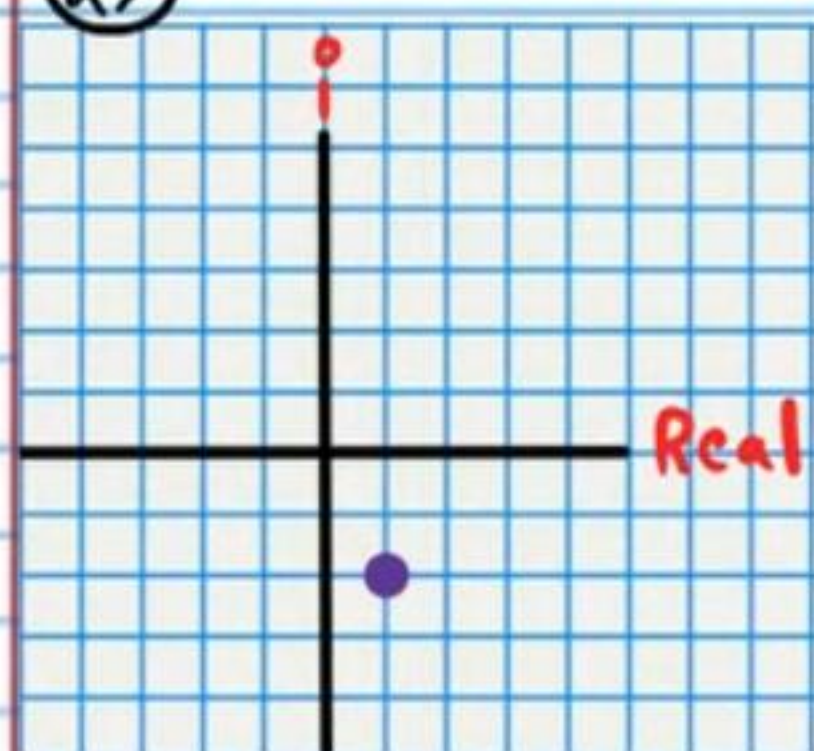
⑬ $4 - 12i$

⑭ $18 + i$

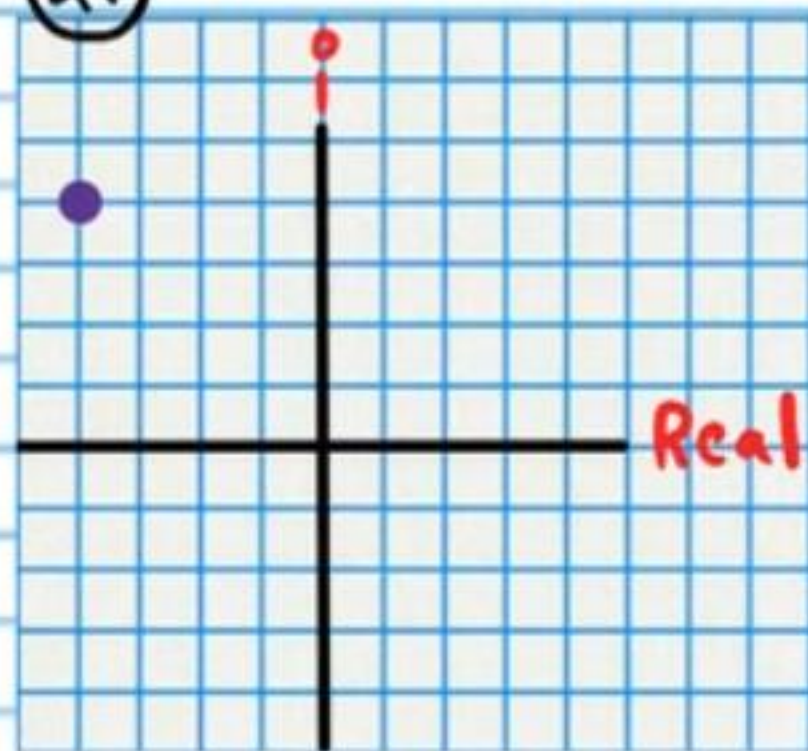
⑮ $4 + 11i$

⑯ $25 - 19i$

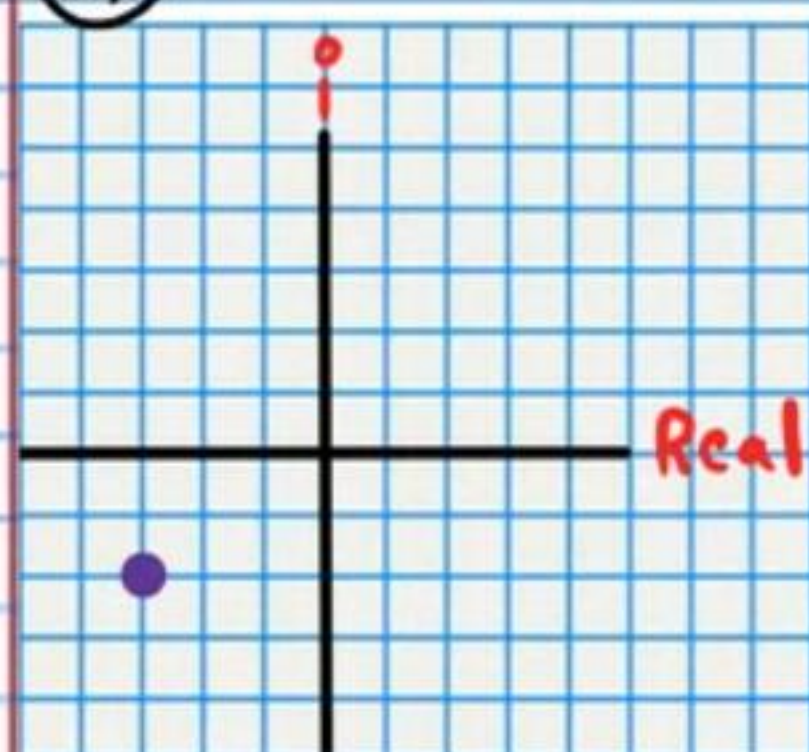
⑰



⑱



⑲



⑳

